

CLARINDA, IA, USA

WMO: 725479

Lat: 40.722N

Lon: 95.026W

Elev: 302

StdP: 97.74

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 04937

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.2	-17.1	-24.0	0.4	-19.8	-21.3	0.6	-15.5	11.9	-2.6	10.9	-3.3	1.8	0	0.507

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.8	34.1	25.0	32.8	24.6	32.0	24.3	27.1	31.8	26.1	30.9	25.2	30.0	4.3	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.0	22.2	30.3	24.9	20.7	29.3	23.7	19.2	28.2	87.7	32.1	82.5	30.8	78.6	29.0	34.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.7	9.1	8.1	DB	-26.2	36.3	4.1	1.7	-29.1	37.5	-31.5	38.5	-33.8	39.5	-36.8	40.7	(4)
(5)			WB	-26.4	28.7	3.6	1.8	-29.0	30.0	-31.1	31.0	-33.1	32.1	-35.7	33.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.9	-4.1	-2.2	4.8	10.9	17.1	22.5	24.3	23.1	19.0	12.0	4.4	-1.7	(6)	
	DBStd	11.37	6.83	6.82	6.87	5.58	4.76	3.47	3.28	3.14	4.55	5.35	5.64	6.48	(7)	
	HDD10.0	1616	437	343	187	54	4	0	0	0	1	41	183	366	(8)	
	HDD18.3	3304	696	576	419	228	81	7	2	3	47	206	418	622	(9)	
	CDD10.0	1952	0	1	27	82	224	374	444	407	271	103	14	4	(10)	
	CDD18.3	598	0	0	1	6	43	132	188	152	66	9	0	1	(11)	
	CDH23.3	6277	0	0	17	109	457	1332	2012	1462	732	133	4	18	(12)	
	CDH26.7	2448	0	0	2	26	142	508	882	578	266	34	1	8	(13)	
(14) Wind	WSAvg	3.2	3.4	3.6	4.0	4.2	3.6	2.9	2.2	2.0	2.5	3.2	3.5	3.3	(14)	
(15) Precipitation	PrecAvg	883	23	26	57	82	123	119	125	103	96	63	51	27	(15)	
	PrecMax	1329	83	78	168	184	248	277	368	386	296	153	149	66	(16)	
	PrecMin	439	0	3	6	17	28	18	16	23	8	6	0	0	(17)	
	PrecStd	182	17	20	37	45	56	58	71	72	61	39	36	17	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.8	18.9	25.2	29.9	33.7	34.2	37.3	35.9	33.9	30.2	23.9	25.1	(19)	
		MCWB	7.9	11.6	15.4	17.9	21.2	24.2	24.8	25.5	23.2	20.1	15.4	22.3	(20)	
	2%	DB	11.1	14.0	22.3	26.4	30.2	32.7	34.9	33.6	32.2	27.2	20.0	13.9	(21)	
		MCWB	6.6	8.3	14.3	16.2	20.3	23.9	26.1	25.4	23.4	18.0	13.9	9.8	(22)	
	5%	DB	7.6	10.2	19.0	23.7	27.7	31.3	32.8	32.2	29.9	24.0	17.2	10.2	(23)	
		MCWB	4.6	6.4	13.0	15.4	18.9	23.2	25.5	25.1	22.2	16.8	11.9	7.2	(24)	
	10%	DB	5.0	7.3	16.1	21.2	26.1	29.9	32.0	30.0	27.6	22.0	14.0	7.3	(25)	
		MCWB	2.5	4.5	11.2	14.1	18.2	22.2	25.0	24.0	20.7	15.7	9.4	4.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.1	12.5	17.8	19.8	24.2	26.8	28.9	28.2	25.8	22.9	17.2	21.9	(27)	
		MCDB	11.7	16.8	22.5	24.9	30.1	32.2	33.7	33.0	31.5	27.5	21.5	25.2	(28)	
	2%	WB	7.1	9.2	15.8	18.2	22.1	25.3	27.6	26.8	24.7	20.6	14.9	11.3	(29)	
		MCDB	10.2	14.1	19.9	24.0	27.9	31.1	32.0	31.5	29.9	23.6	18.3	12.9	(30)	
	5%	WB	4.9	6.7	13.9	16.6	20.7	24.3	26.5	25.7	23.5	18.5	12.6	7.3	(31)	
		MCDB	8.0	9.9	17.9	22.4	25.7	29.7	31.2	30.3	27.8	22.1	16.8	9.5	(32)	
	10%	WB	2.8	4.6	11.4	15.0	19.5	23.3	25.4	24.8	22.2	16.2	10.0	4.8	(33)	
		MCDB	4.8	7.3	16.4	19.9	24.1	28.3	30.0	29.2	26.4	20.4	13.9	7.4	(34)	
(35) Mean Daily Temperature Range	MDBR		10.7	10.9	12.4	13.3	12.0	11.6	11.8	11.8	13.9	13.6	12.2	10.3	(35)	
	5% DB	MCDBR	15.1	15.8	17.7	17.7	14.6	13.0	13.3	13.7	15.5	16.9	17.0	14.4	(36)	
		MCWBR	11.1	10.7	10.3	9.7	6.8	6.0	6.3	7.0	7.6	9.1	10.8	10.8	(37)	
	5% WB	MCDBR	14.5	14.8	14.1	14.6	11.9	11.6	11.6	11.9	13.0	13.3	14.9	13.0	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.282	0.315	0.327	0.384	0.411	0.417	0.430	0.422	0.381	0.335	0.296	0.285	(40)	
	taud		2.456	2.376	2.409	2.272	2.265	2.293	2.247	2.287	2.357	2.471	2.514	2.463	(41)	
	Ebn at Noon		886	898	926	888	868	860	845	841	856	865	863	852	(42)	
	Edh at Noon		77	97	105	130	134	130	135	127	110	88	73	71	(43)	
(44) All-Sky Solar Radiation	RadAvg		2.06	2.97	3.86	4.87	5.77	6.40	6.43	5.61	4.69	3.25	2.26	1.74	(44)	
	RadStd		0.17	0.19	0.35	0.30	0.48	0.53	0.46	0.31	0.31	0.37	0.18	0.18	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (44 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page